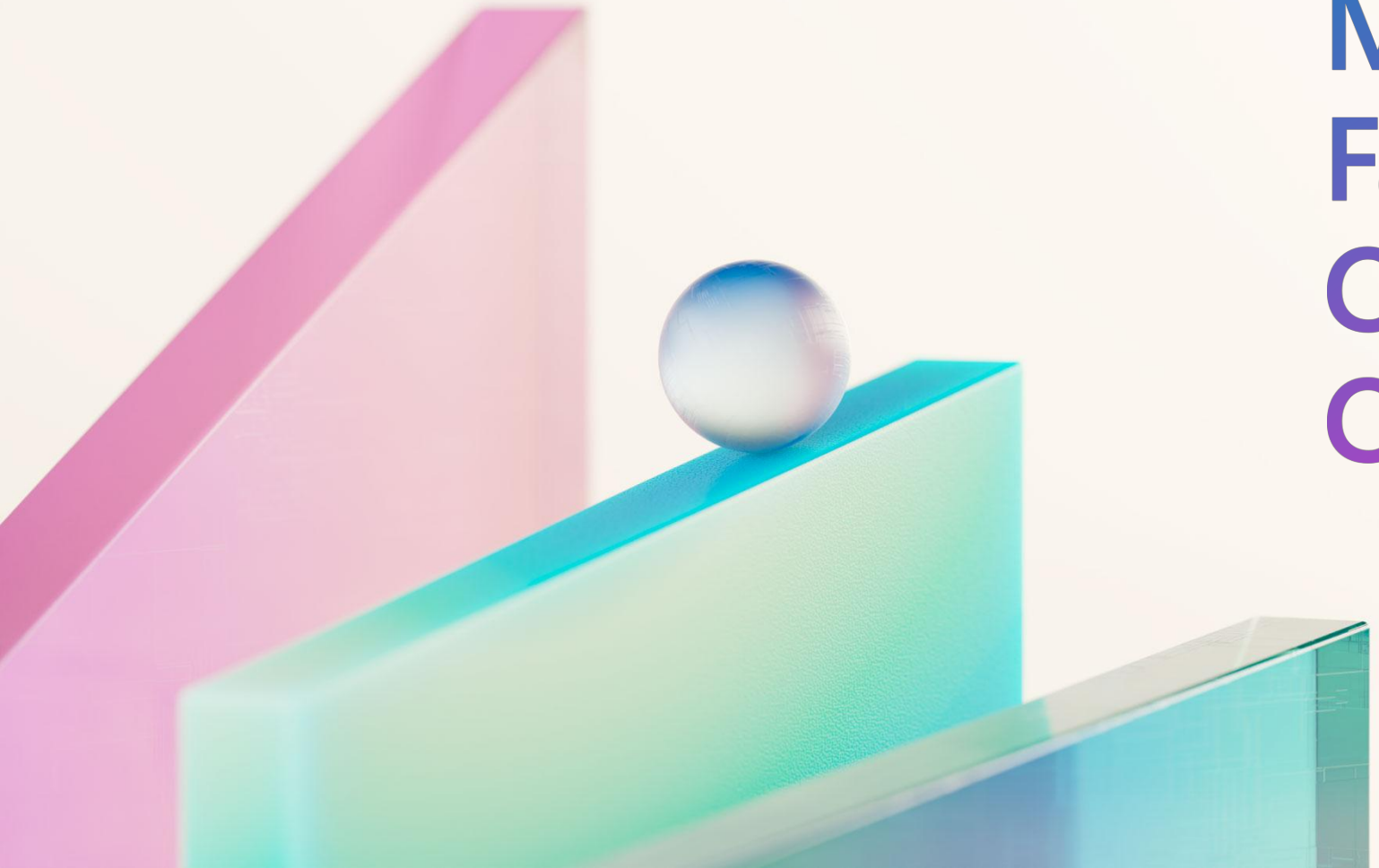
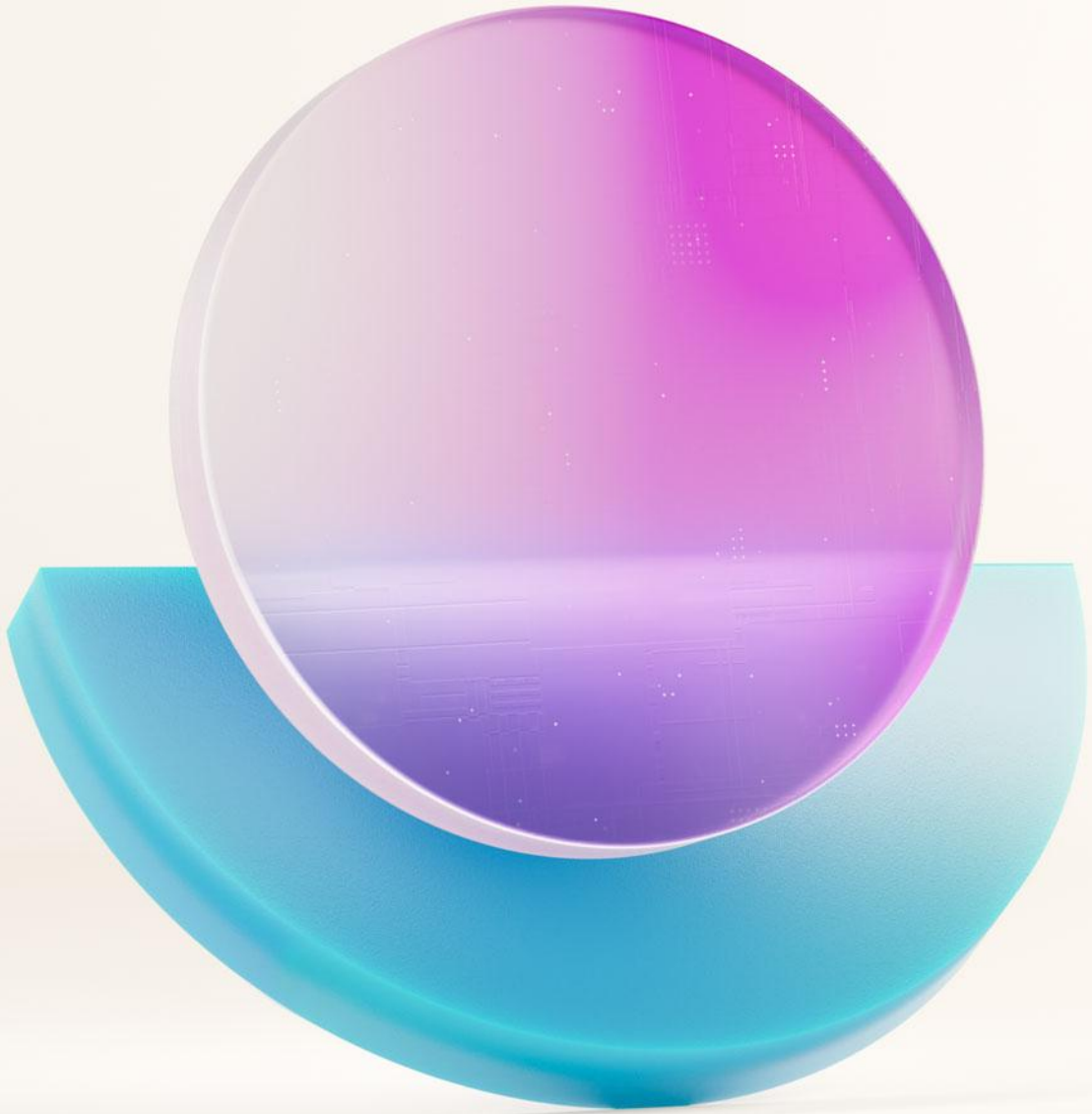
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aka.ms/FabricCommunity
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aka.ms/FabricUserGroups
Find a user group that matches your interests in your area or online



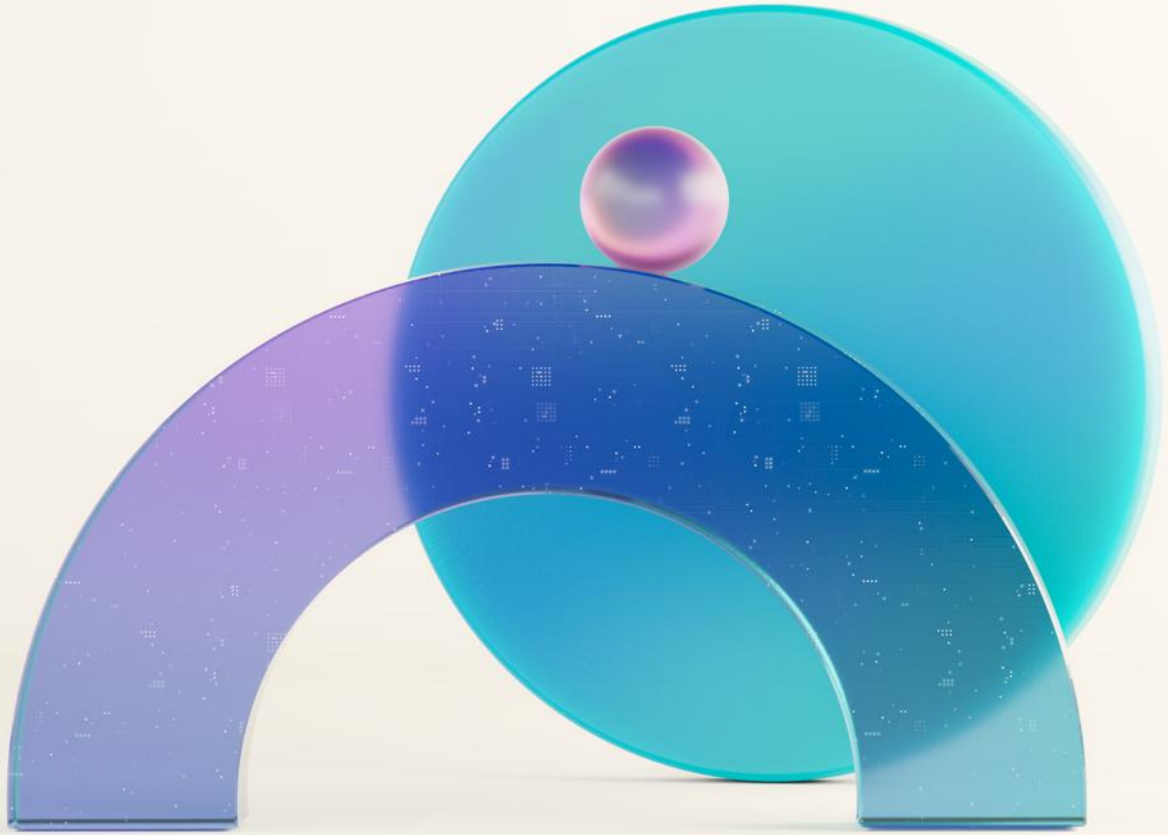
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Adopting Microsoft Fabric: Data Mesh and Medallion Architecture Made Easy

Harry Arce
Data, Apps & AI Technical Specialist
Microsoft

https://hihello.me/hi/harryfabcon

Agenda

01

Microsoft Fabric Overview

02

Data Architecture Challenges

03

Vertical Approach: Data Mesh

04

Horizontal Approach: Medallion Architecture

05

Summary

07

Q & A

AI is transforming the world

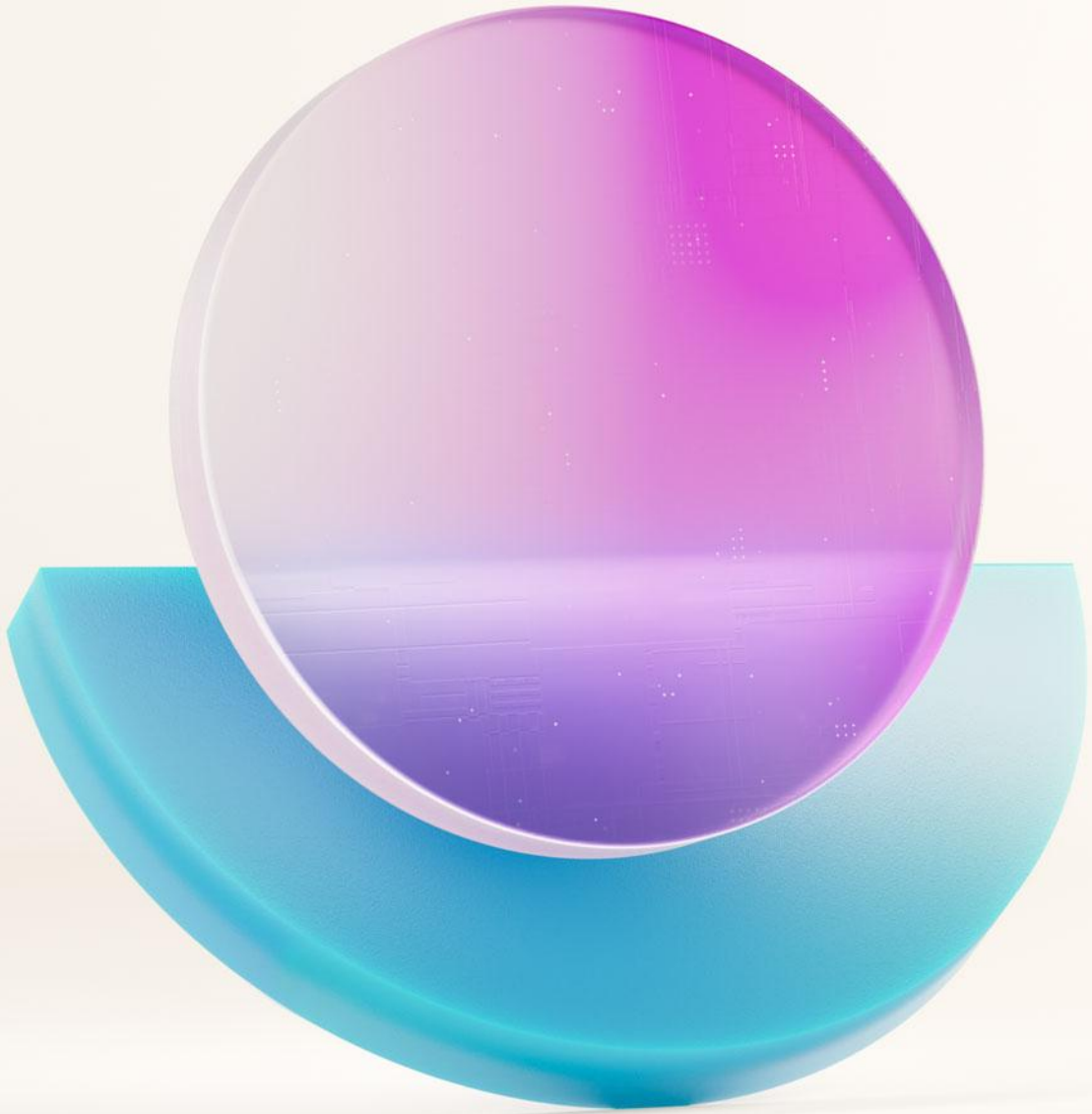
Data is the fuel that powers AI

But AI is only as good as your data architecture

To reap the benefits of AI-powered tools, businesses need a modernized data and analytics estate

Siloed and fragmented tech stack Disconnected systems and data silos	**Unified, connected single source of truth** Integrated systems for consistent data flow
Data Underutilization Poor accessibility hindering AI adoption	**Accessible data and AI solutions** User-friendly and customizable solutions
Limited collaboration and scalability Restricted sharing capabilities	**Enhanced visibility and decision-making** Timely access to insights for empowered decisions

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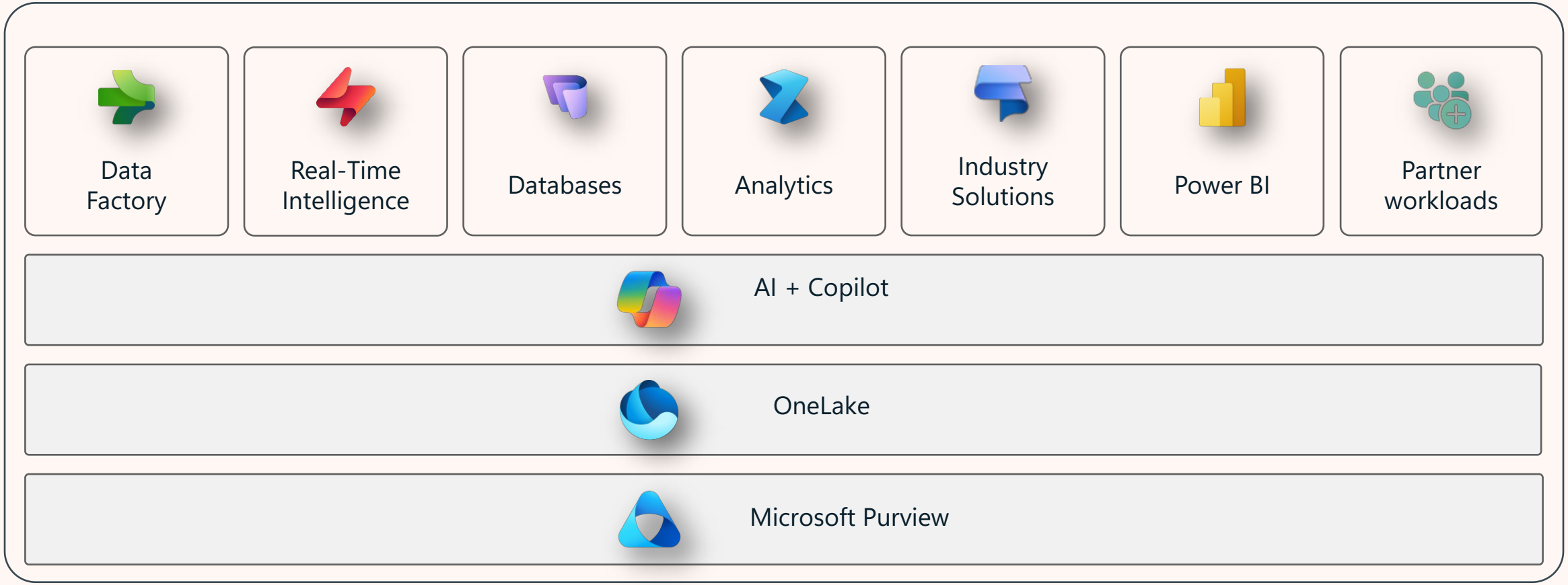


Microsoft Fabric

The unified data platform for the era of AI

Microsoft Fabric

The unified data platform for the era of AI



Microsoft Fabric

End-to-end analytics data fabric from the data lake to the business user

Complete Analytics Platform

- Everything, Best of Breed
- Unified
- Copilots

Lake centric and open

- OneLake
- One Copy
- Always Synced

Empower Every Business User

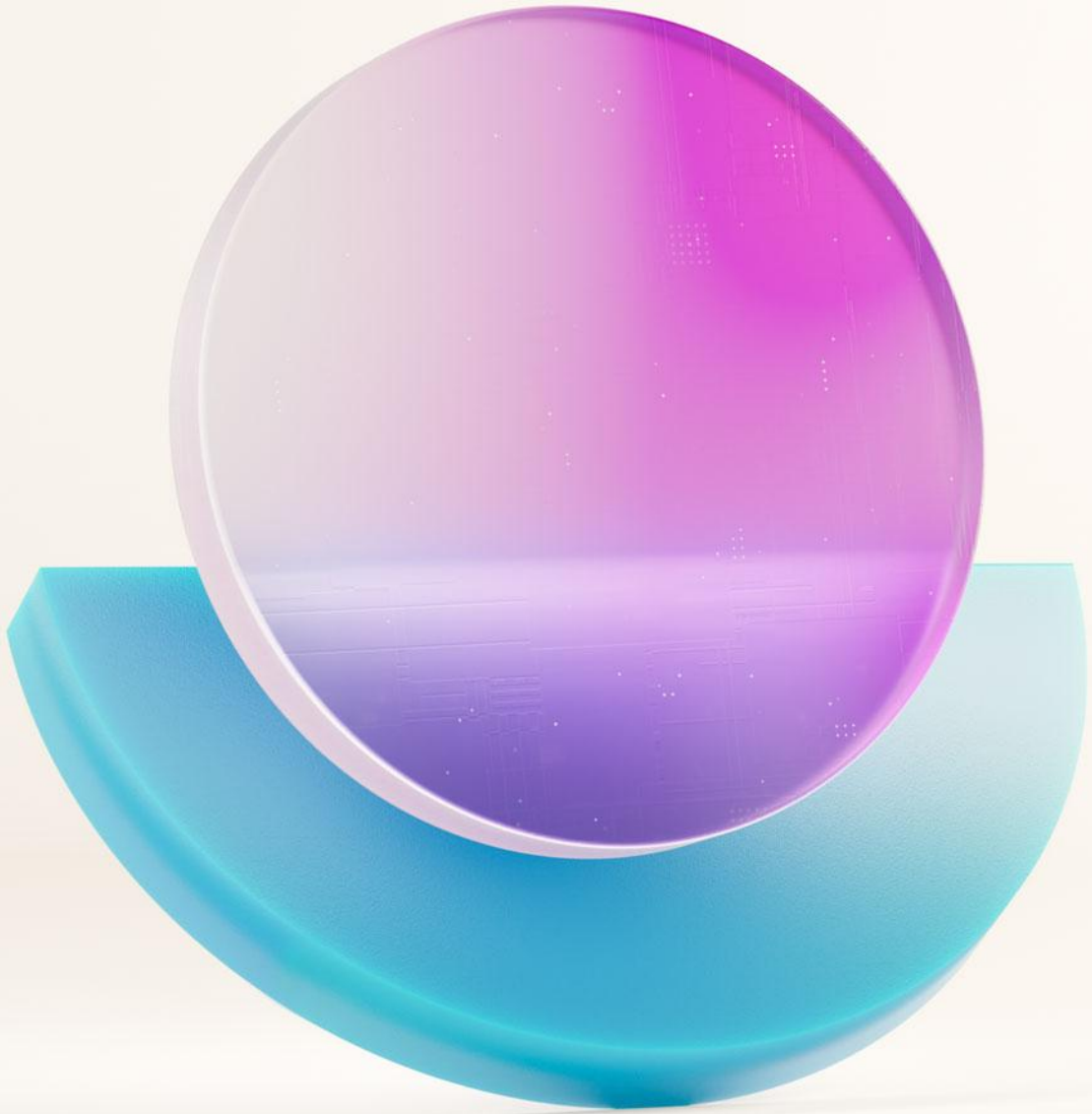
- Familiar and intuitive
- Built into Office
- Insight to action

Pervasive security and governance

- End to end visibility
- Always governed
- Secure by default

AI Powered

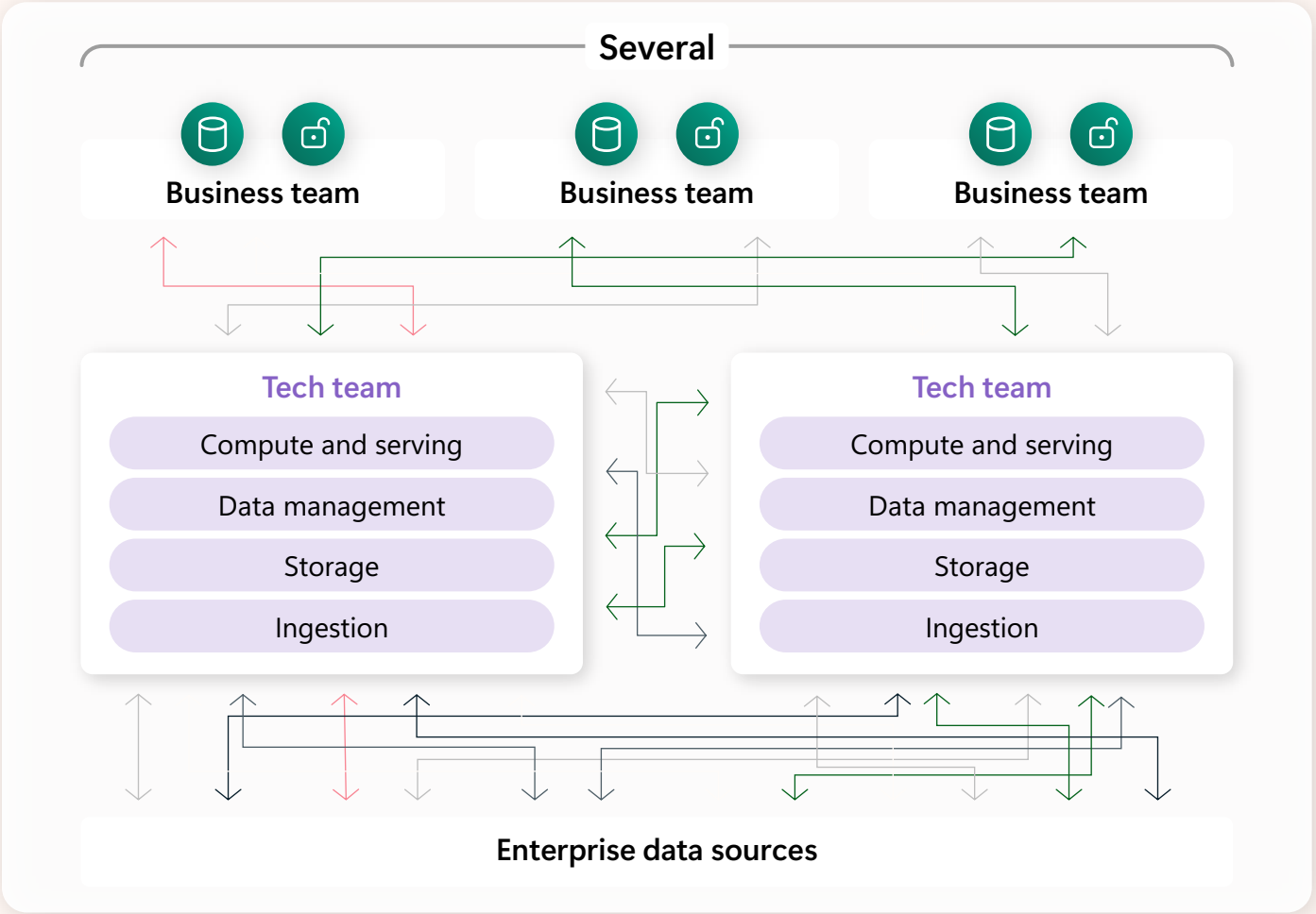
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Data Architecture Challenges

The starting line:

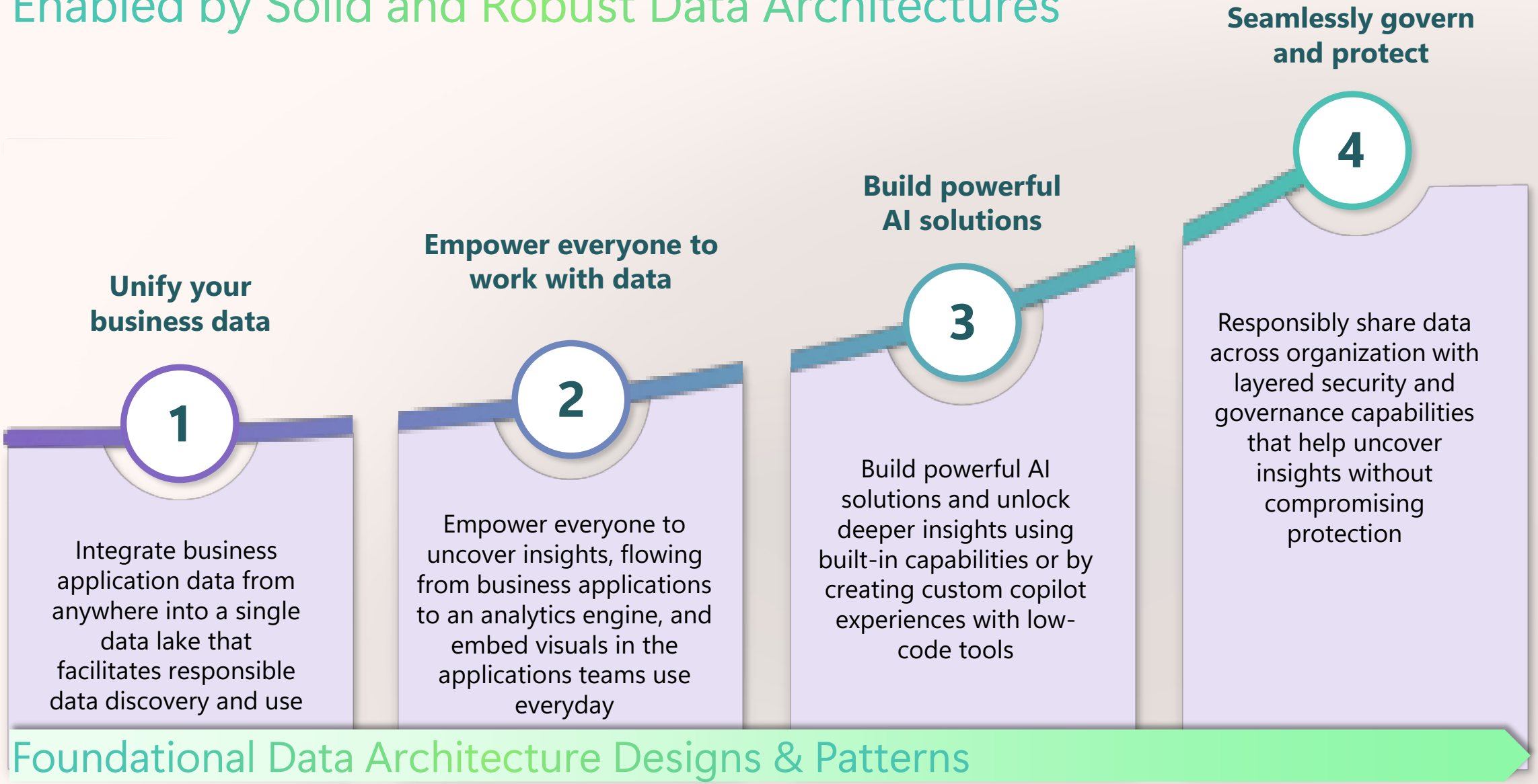
A complex, organically evolved data estate



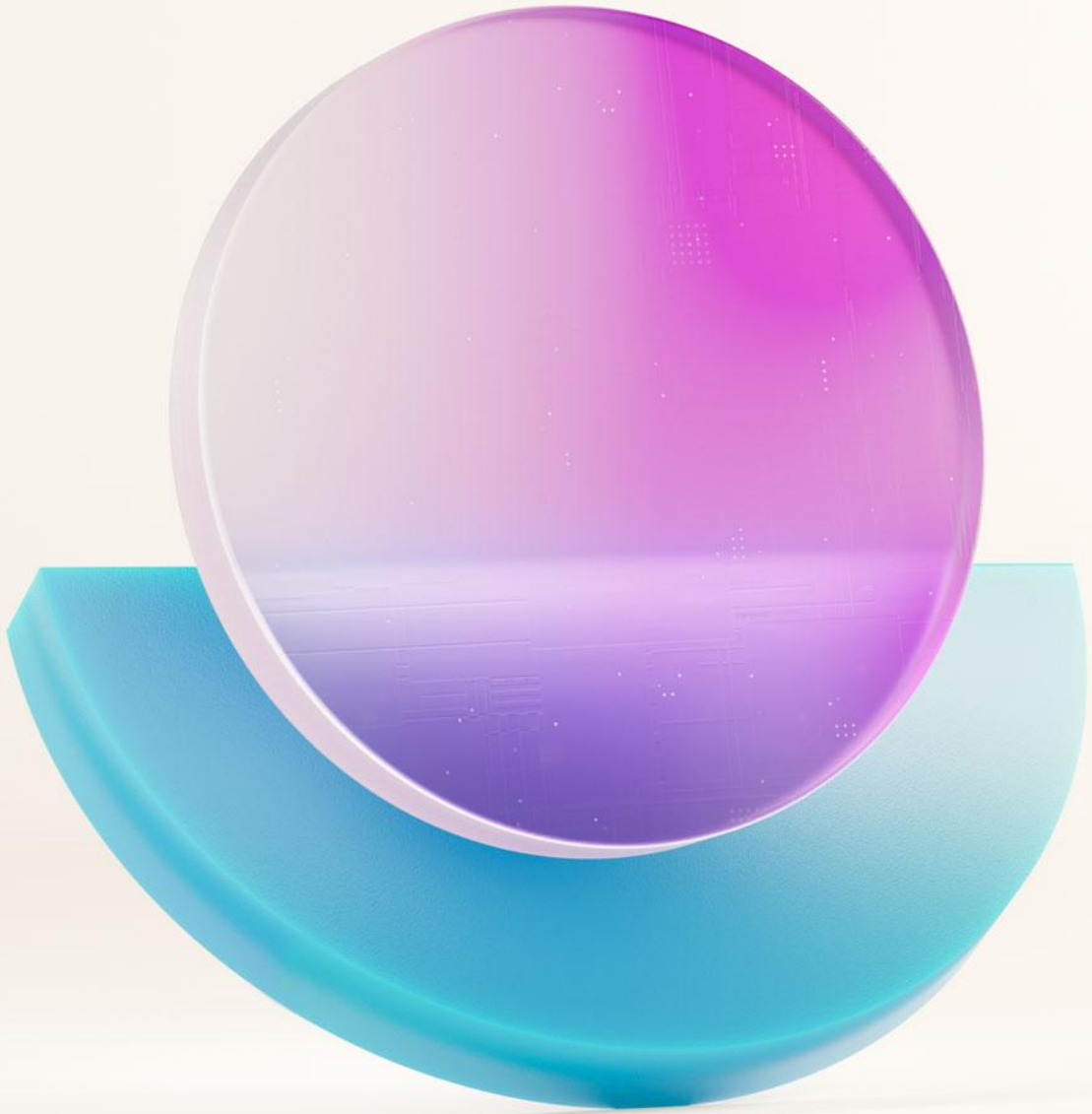
1. Data copies and infrastructure inefficiencies
2. Limited interoperability and collaboration across teams
3. Data exposure risks

Successful Data, AI, and App Strategies

Enabled by Solid and Robust Data Architectures



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Vertical Approach to Data Architecture Decomposition:

Data Mesh in Microsoft Fabric

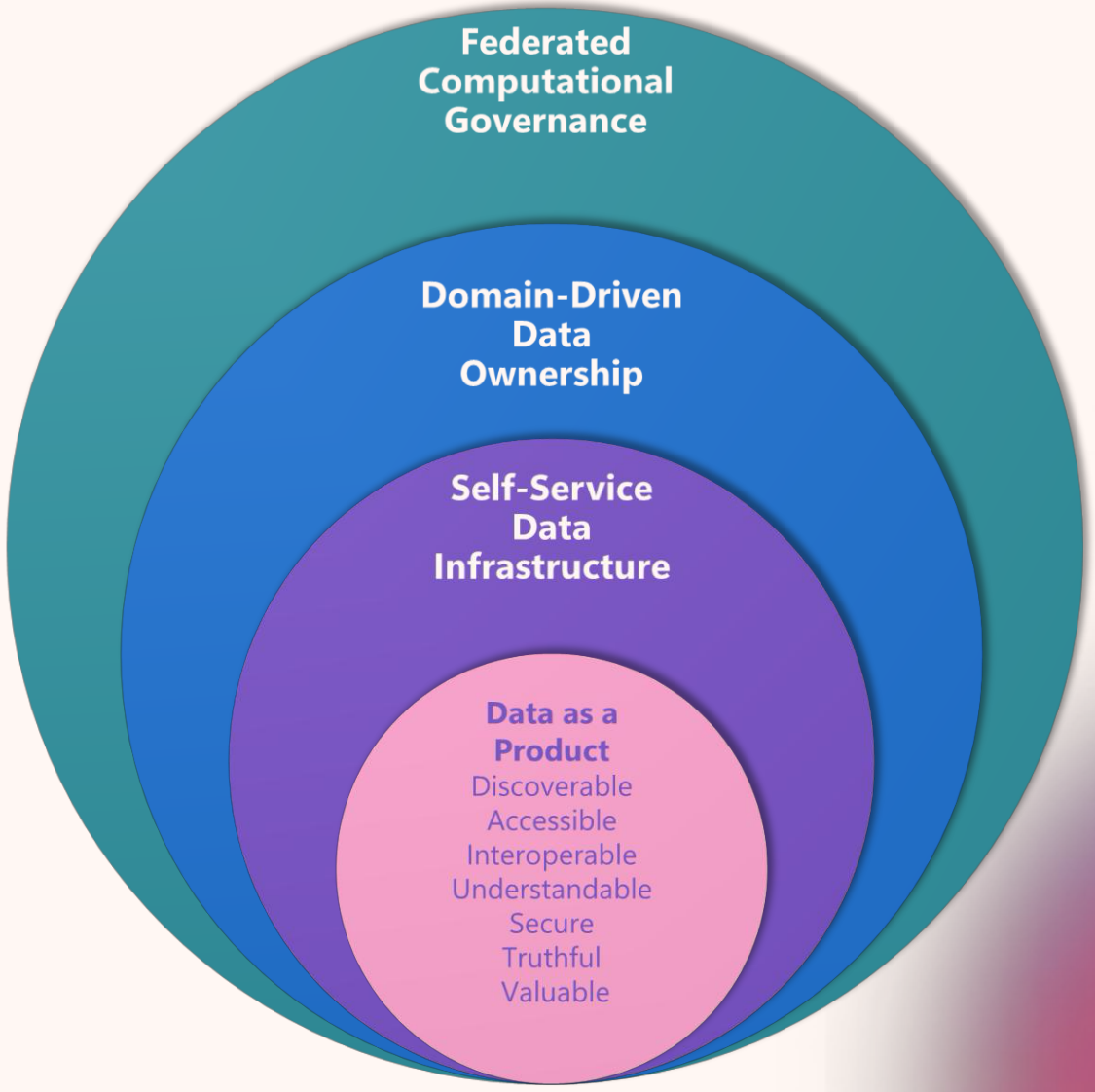
Data Mesh: A vertical approach to data architectures

Architectural pattern inherited from Domain Driven Development (DDD) approach for implementing **decentralized** enterprise data platforms.

Centralized Systems Monolithic data solutions	**Domain-driven data solutions** Smaller and manageable data products
Centralized Team Platform provider and Data integrator	**Autonomous teams of experts** Domain teams produce and consume data
Limited Scalability and Delivery Backlogs and delays in data solutions delivery	**Enhanced innovation and decision-making** Accelerated data to insights journey

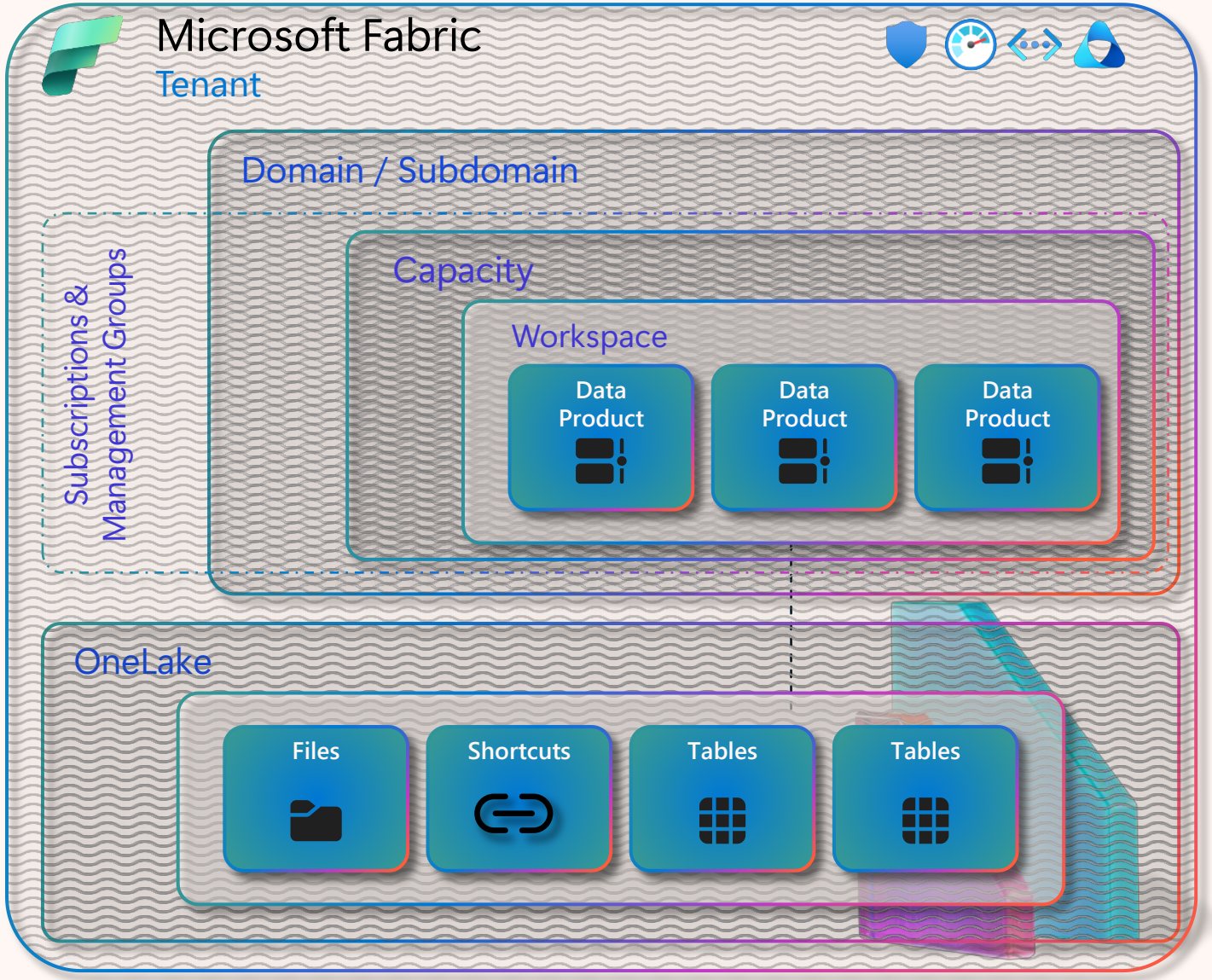
Data Mesh: A vertical approach to data architectures

Key Principles

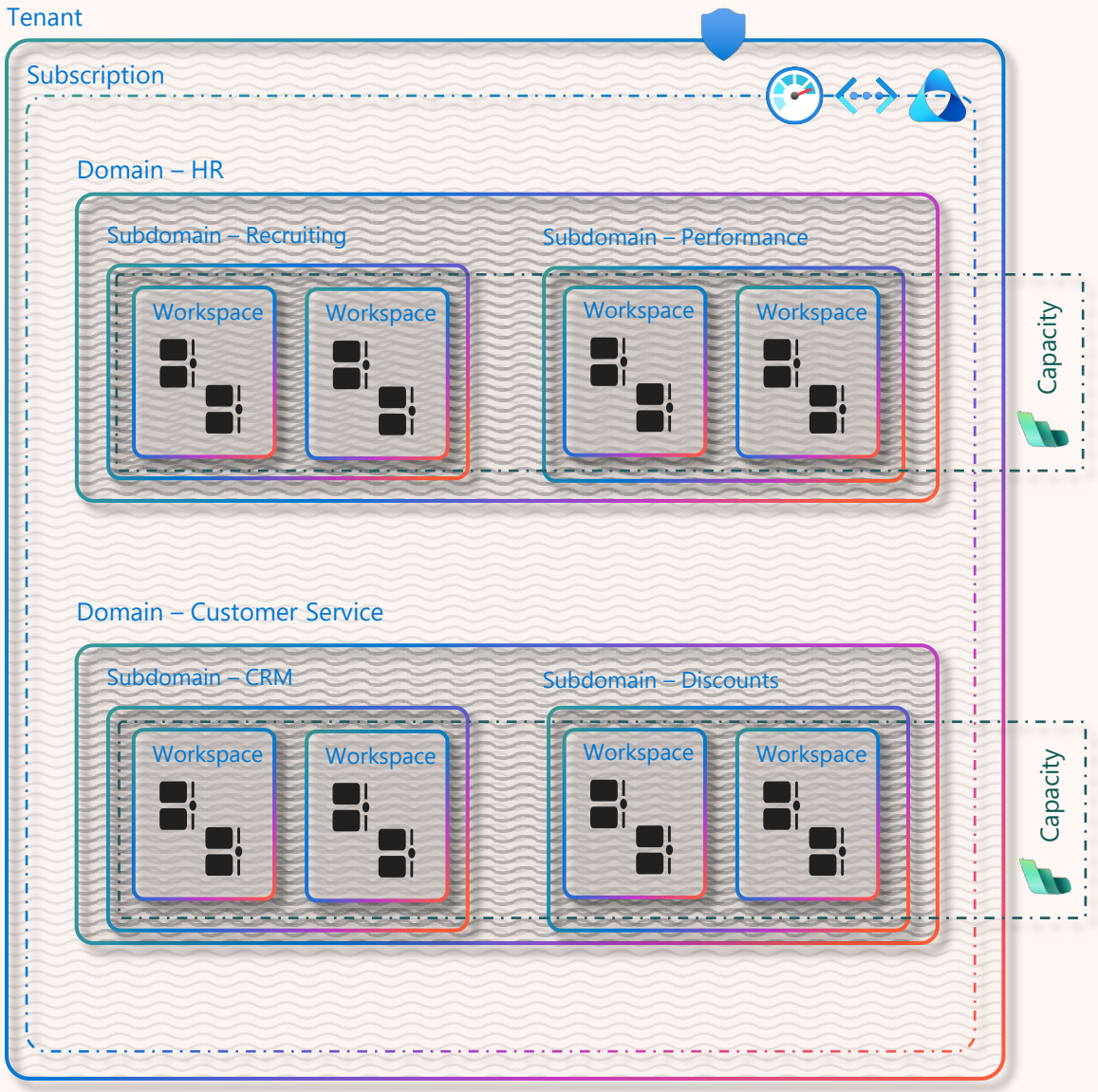


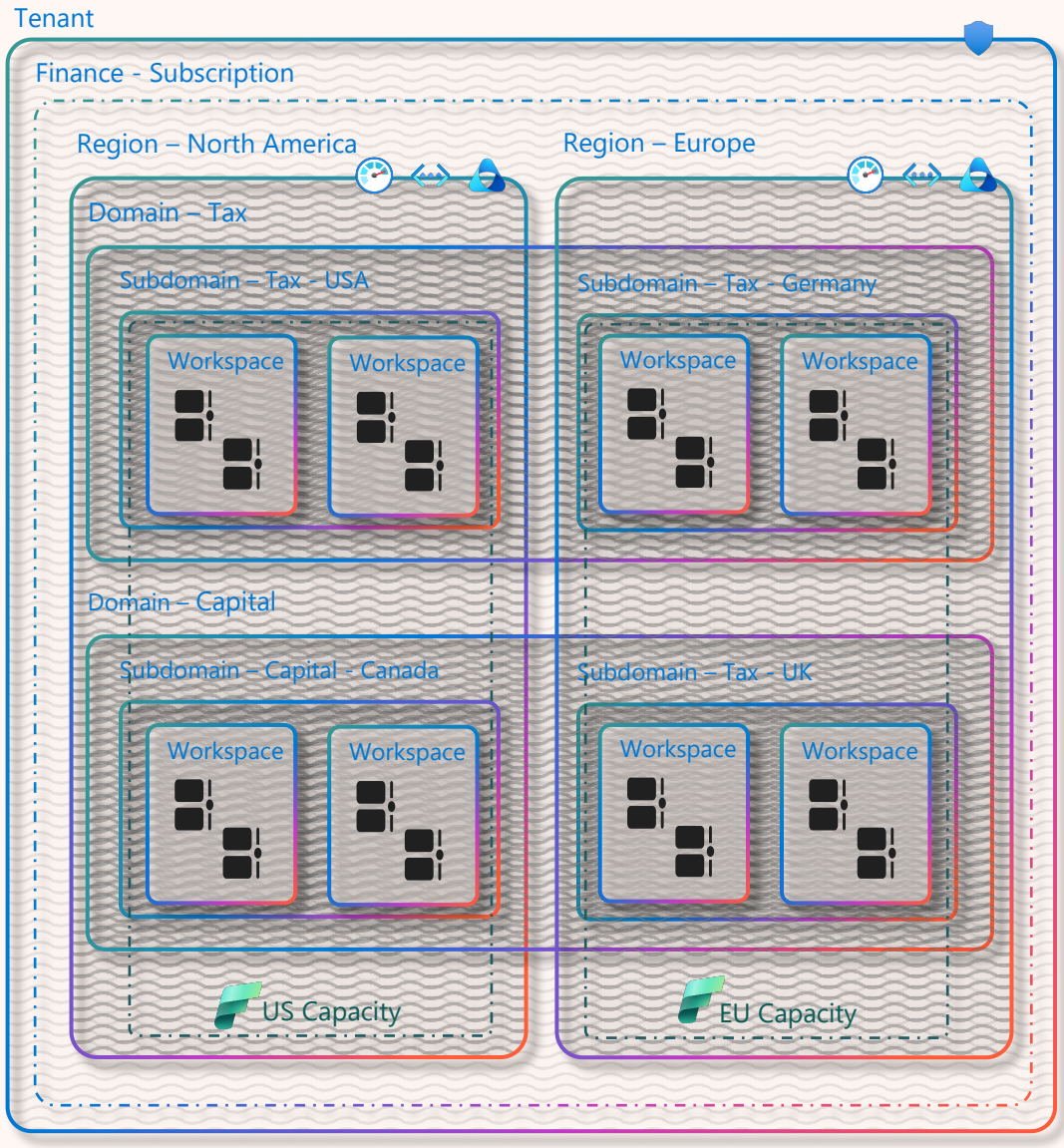
Data Mesh in Microsoft Fabric

- Federated Computational Governance
- Domain-Driven Data Ownership
- Self-Service Data Infrastructure
- Data as a Product



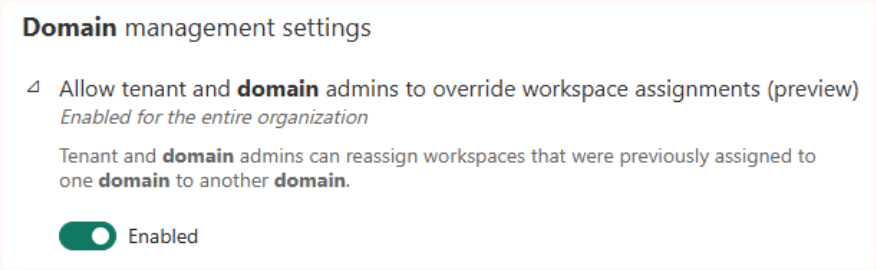
Data Mesh Patterns in Microsoft Fabric

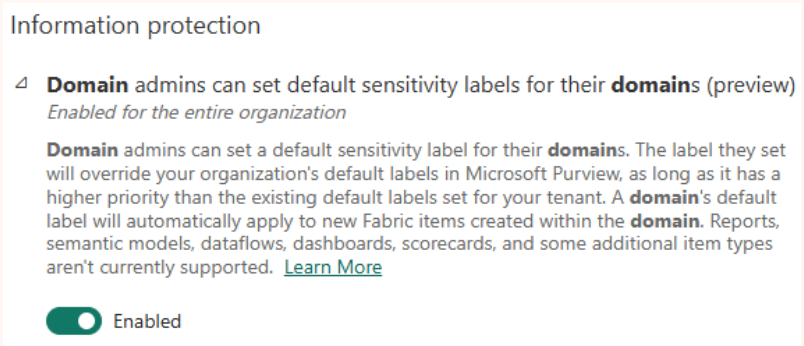




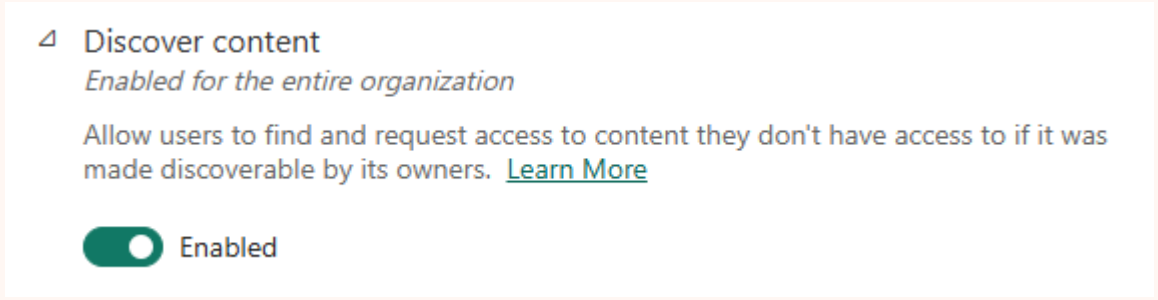
Data Mesh settings in Microsoft Fabric

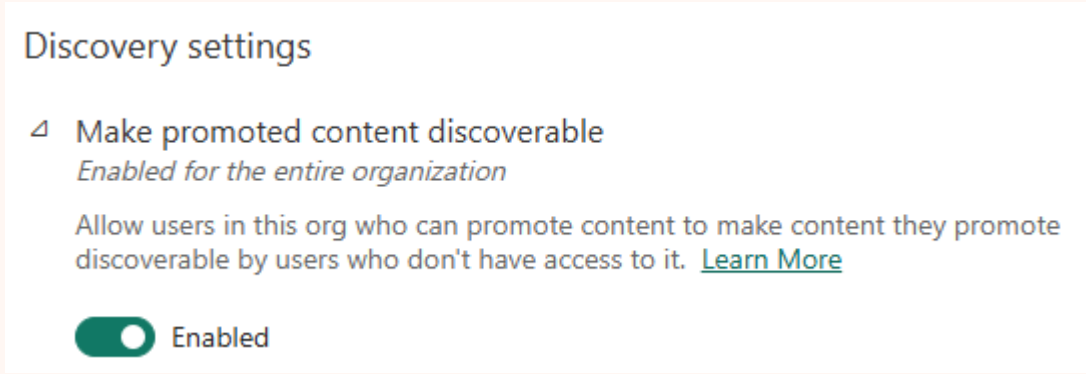
Domain Management

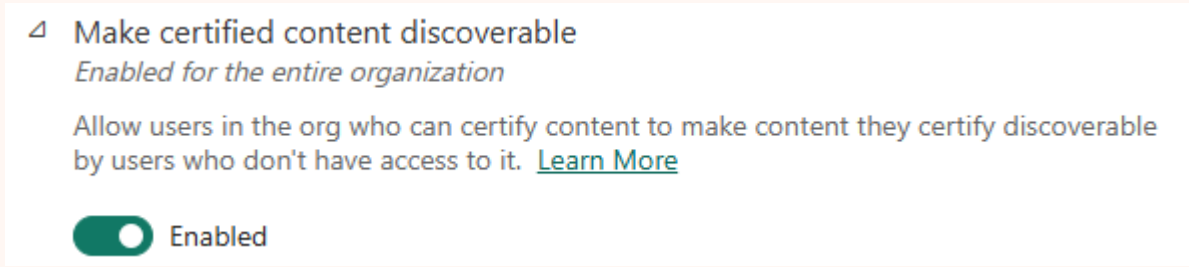




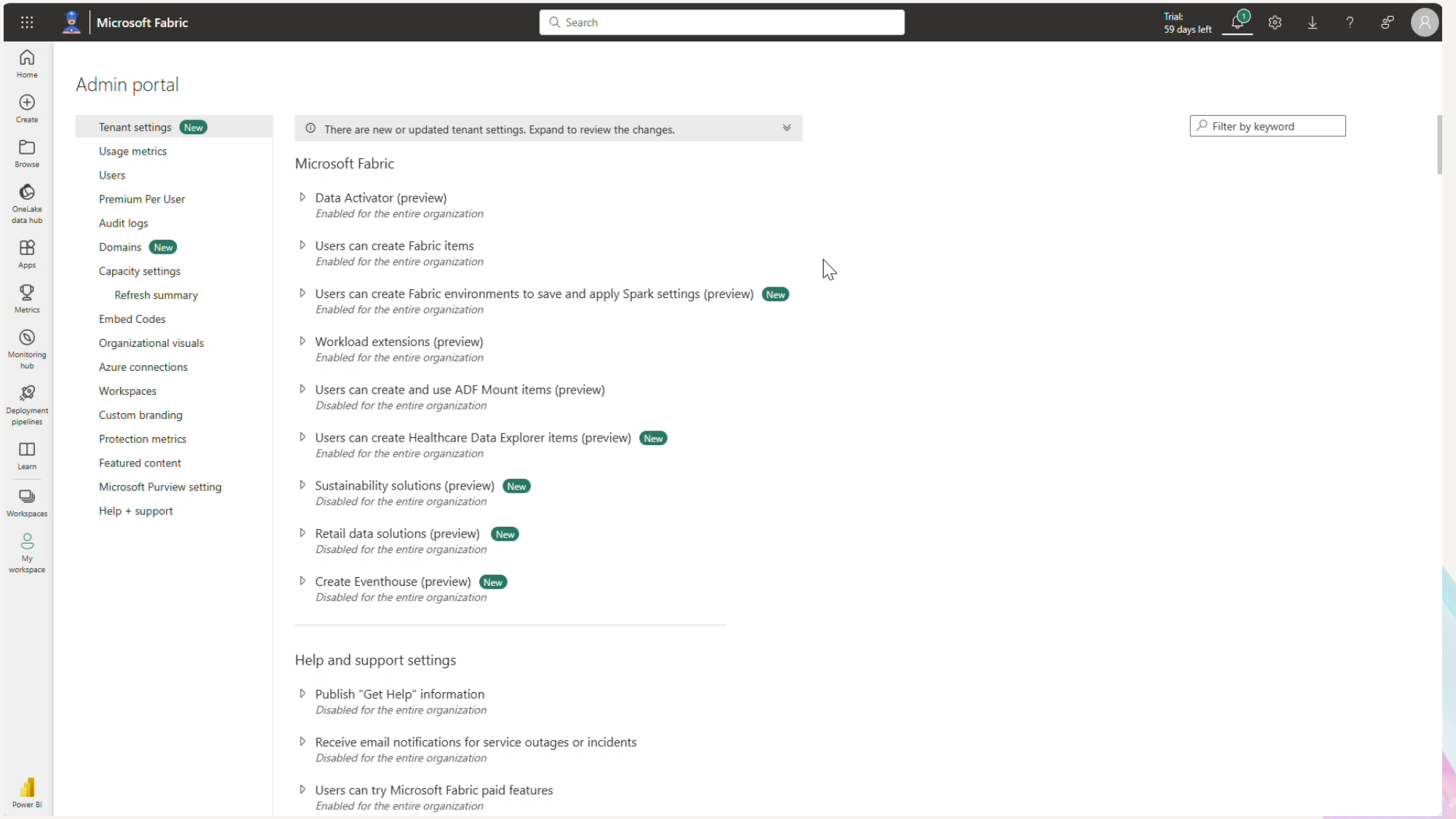
Data Discoverability





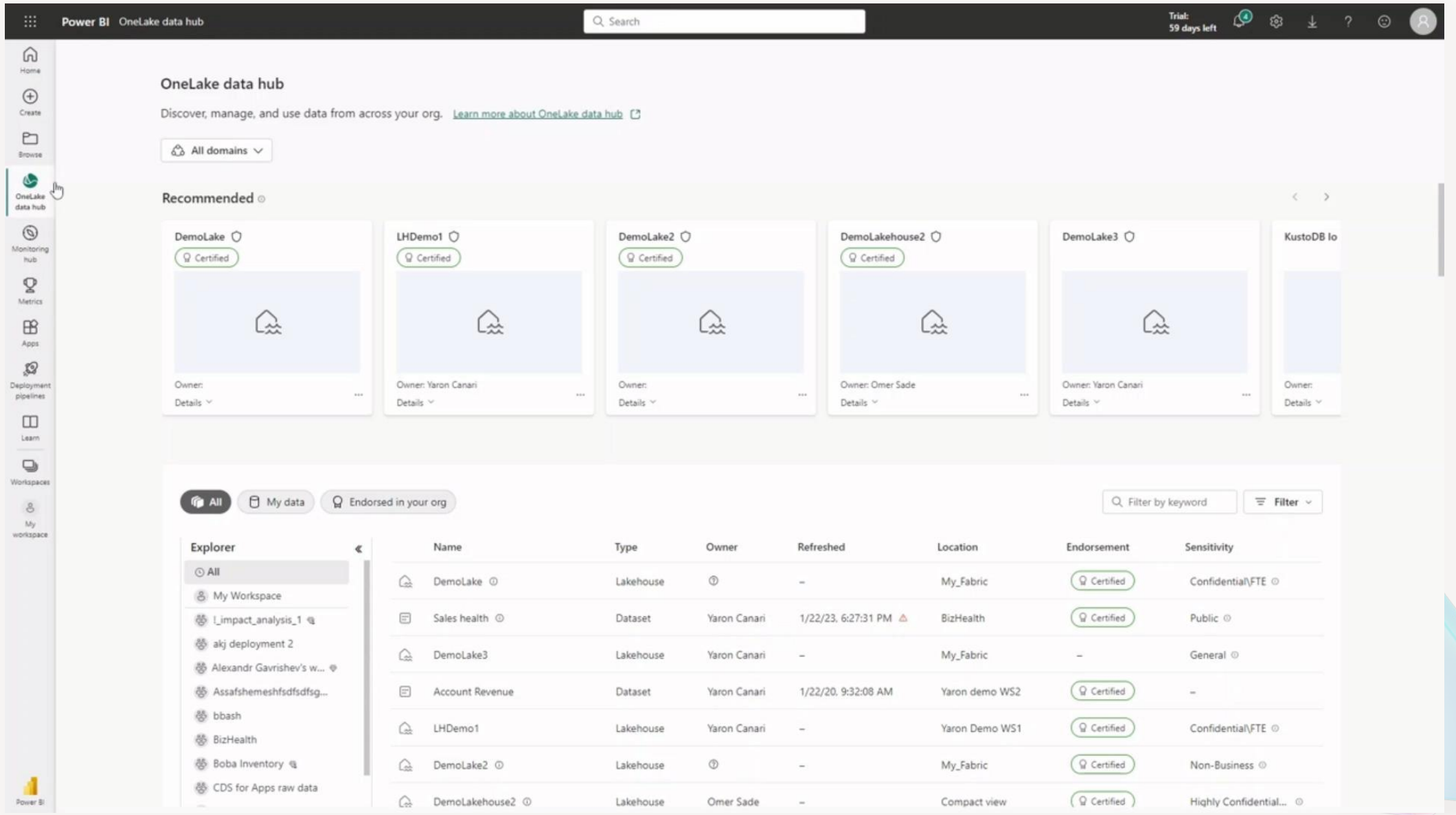


Domains in Microsoft Fabric



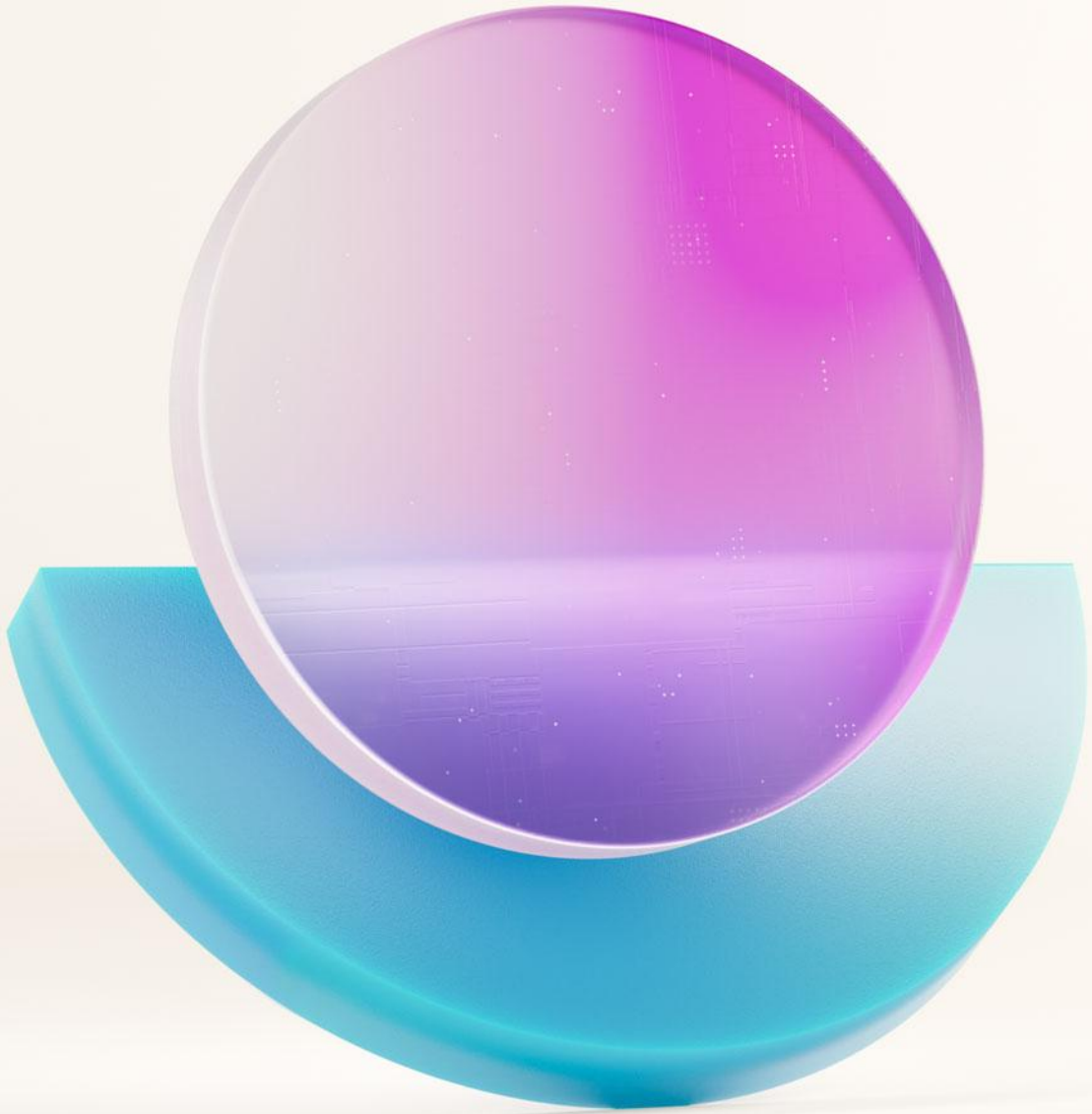
video

OneLake Data Hub in Microsoft Fabric



video

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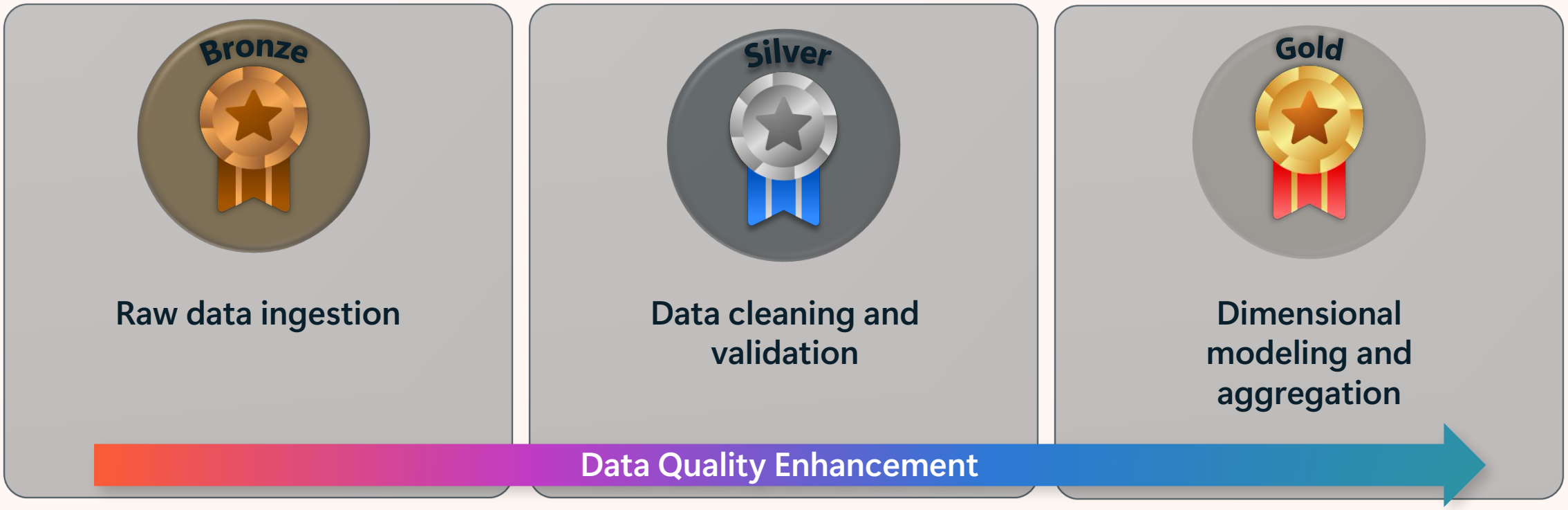
Horizontal Approach to Data Architecture Decomposition:

Medallion Architecture in Microsoft Fabric

Medallion: A horizontal approach to data architectures

The **Medallion Architecture** is a multi-layered approach to building and delivering **data products**. It incrementally and progressively enhances **data quality**, making it ideal for business intelligence (BI) and artificial intelligence (AI) applications.

A Common Pattern



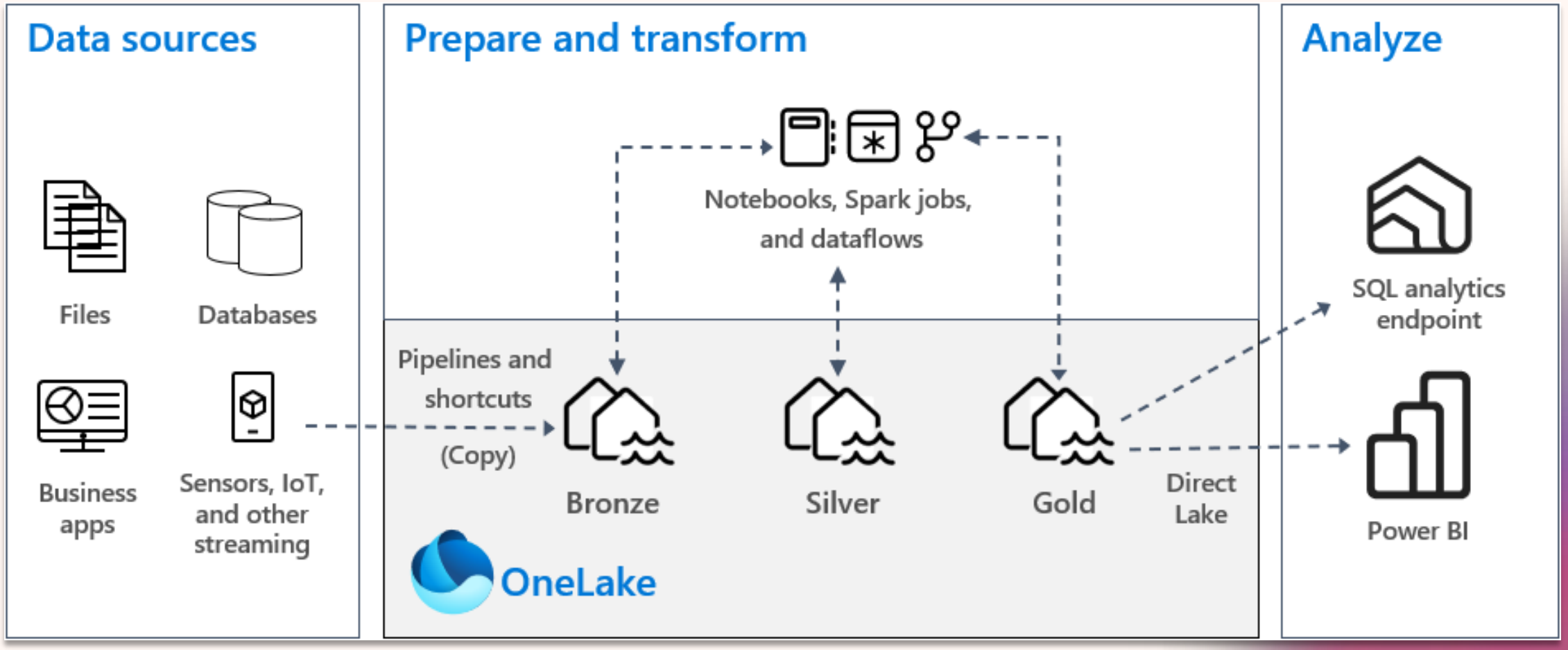
Medallion Architecture

			
Intended Users	• Data engineers • Data operations • Compliance and audit teams	• Data engineers • Data analysts • Data scientists	• Business analysts • Decision makers • BI Developers • Data scientists • ML engineers
Operations	• Minimal to no transformations • Source format • No data cleanup • No validations • Append-only	• Schema enforcement • Data validation • Quality checks (null/missing) • Aggregations • Type Casting	• Tailored for analytics and reporting • Aligned semantically to business functions • Optimized for query performance • Aggregated

Data Quality Enhancement

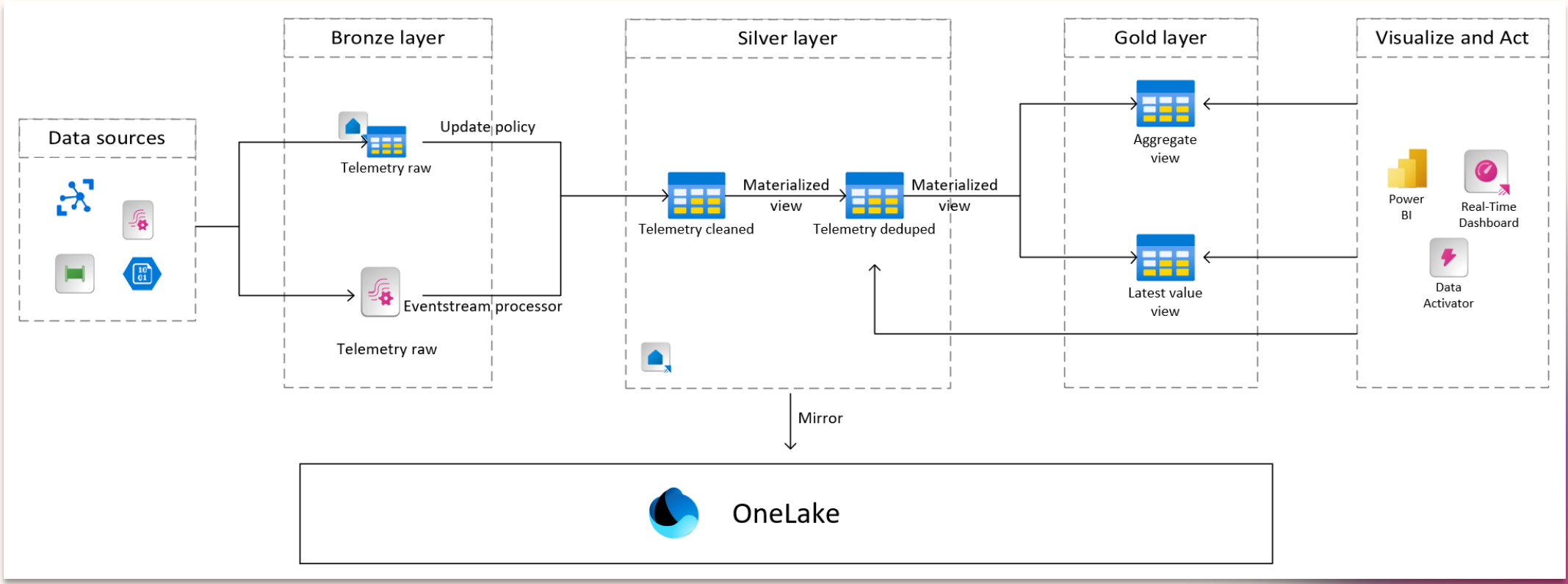
Medallion in Microsoft Fabric

Medallion Architecture Using Lakehouses



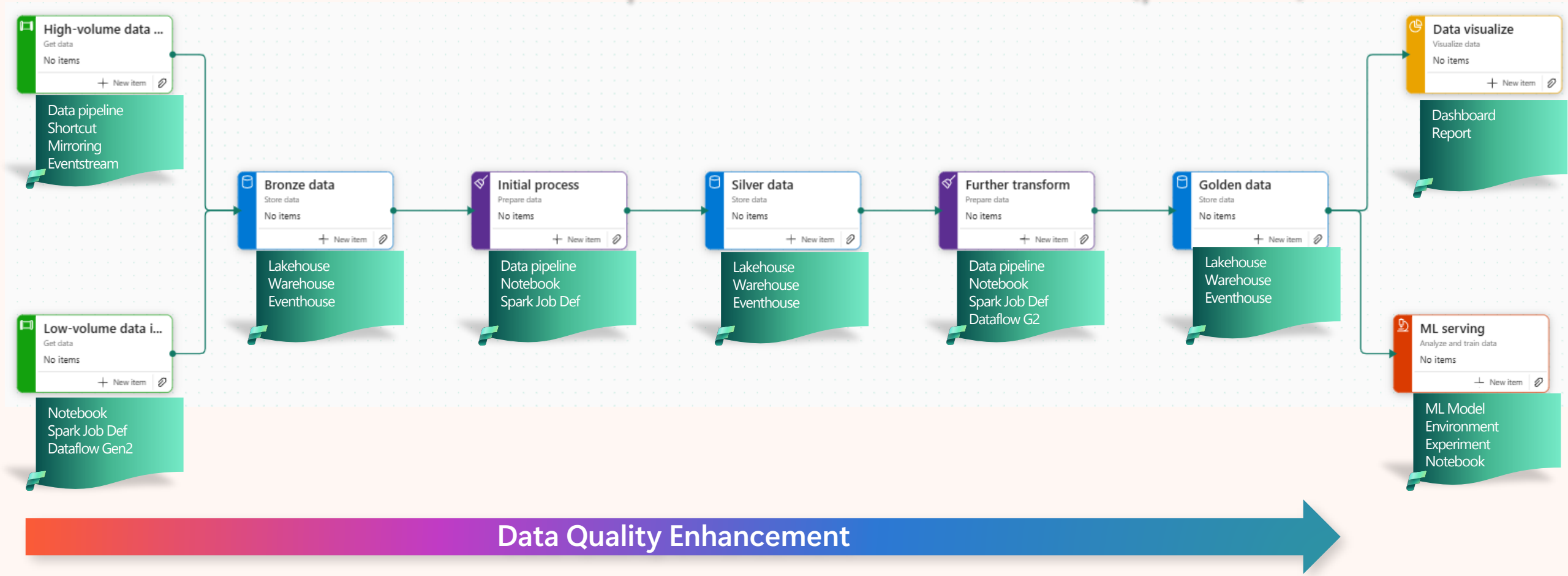
Medallion in Microsoft Fabric

Medallion Architecture in Real-Time Intelligence



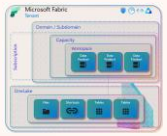
Medallion Visualization in Microsoft Fabric

End-to-End Workspace View with Task flow (preview)



Summary

Data Mesh



Benefits

- Specialization
- Democratization
- Scalability

Challenges

- Organizational Change
- Orchestration
- Standardization

Medallion



Benefits

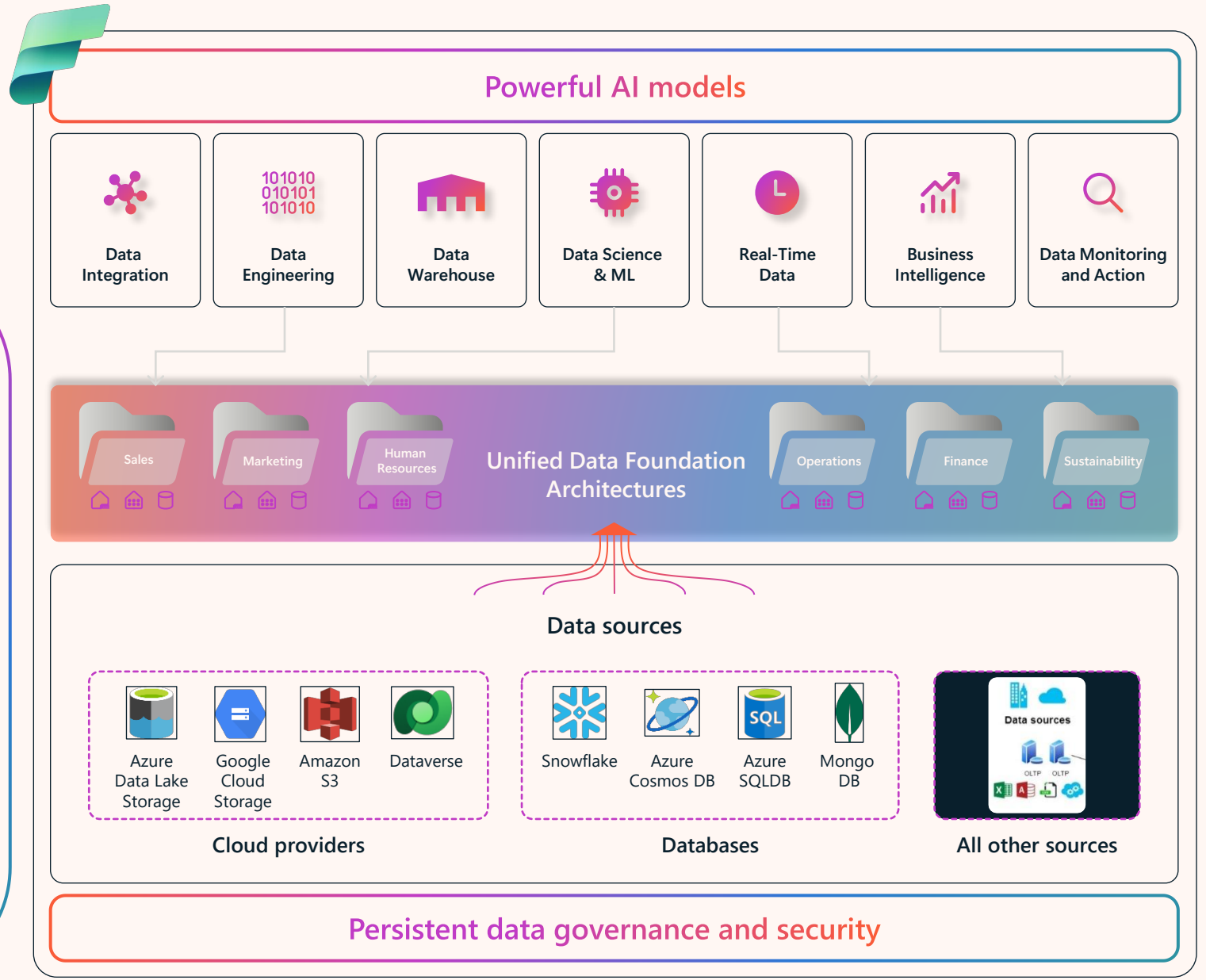
- Enhanced Data Quality
- Modularity & Scalability
- Data Governance & Lineage

Challenges

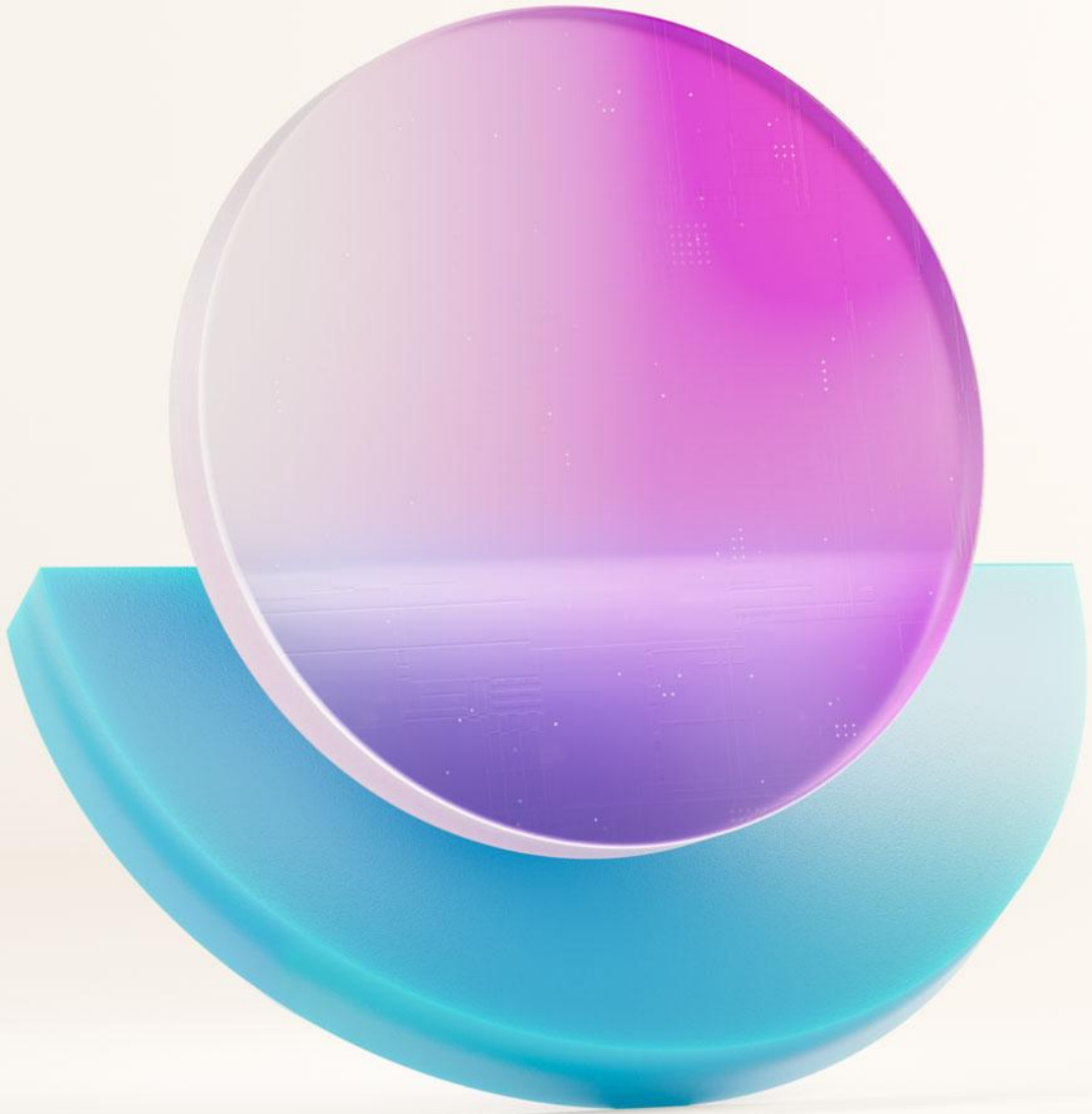
- Maintenance
- Latency
- Complexity

The finish line:

A **unified, intelligent** *data estate* built on an **open and governed** *data platform* with **strong architectural designs and patterns** to meet demands of modern analytics & AI needs



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Q & A

https://hihello.me/hi/harryfabcon

Training Resources & Accelerators

Resources

Data Mesh

- What is a data mesh? - Cloud Adoption Framework | Microsoft Learn
- Establishing Data Mesh architectural pattern with Domains and OneLake on Microsoft Fabric | Microsoft Community Hub
- Easily implement data mesh architecture with domains in Fabric | Microsoft Fabric Blog | Microsoft Fabric
- Use Microsoft Purview to govern Microsoft Fabric - Microsoft Fabric | Microsoft Learn
- Governance domains in Microsoft Purview | Microsoft Learn

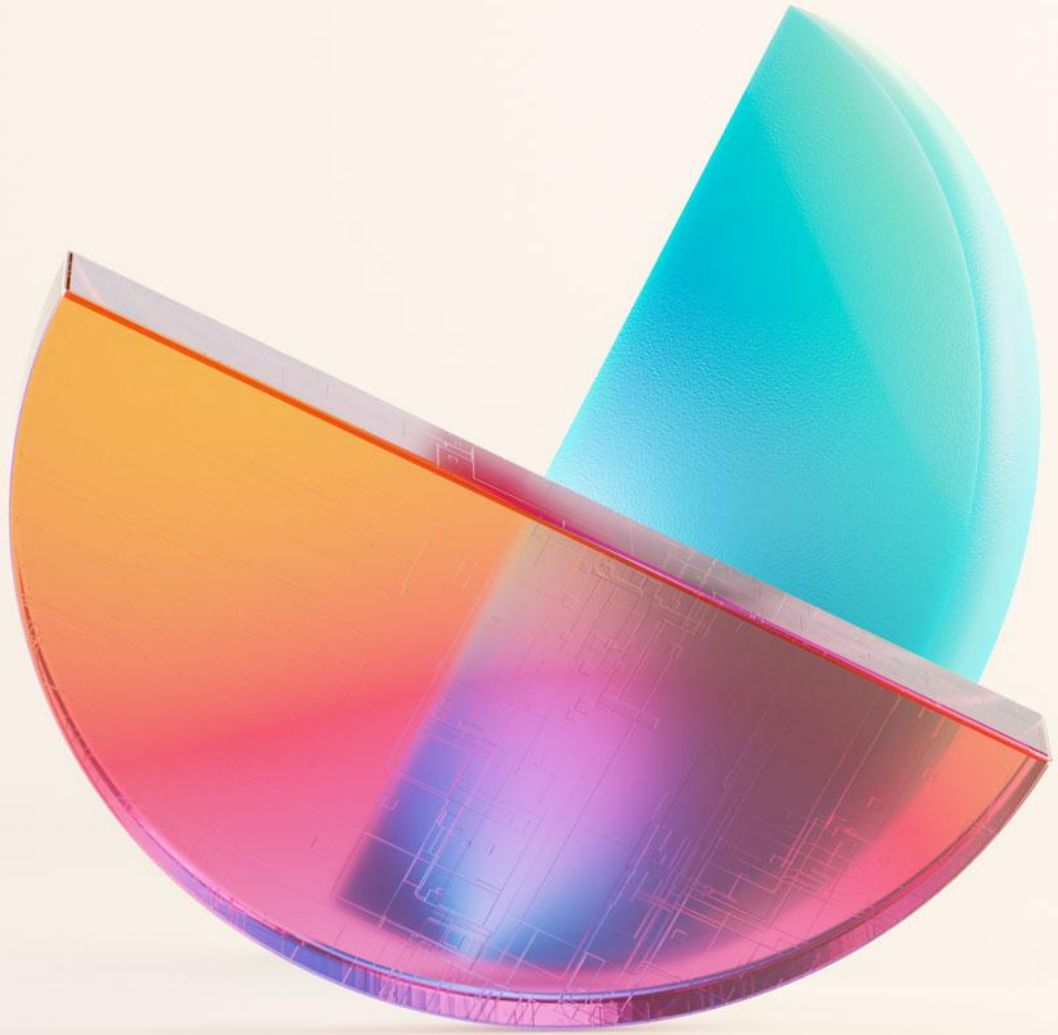
Medallion Architecture

- Implement medallion lakehouse architecture in Fabric - Microsoft Fabric | Microsoft Learn
- Implement medallion architecture in Real-Time Intelligence - Microsoft Fabric | Microsoft Learn
- Organize a Fabric lakehouse using medallion architecture design - Training | Microsoft Learn
- Introducing the Fabric Accelerator! | LinkedIn



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Learn more about Microsoft Fabric

Power your AI transformation with a complete data platform